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Exercise 4.2. Suppose a fair die is rolled twice and that X is the absolute value of the difference of the two rolls. Find the pmf and the CDF of X and plot the CDF. Find a median of X . Is the median unique? *Yes*

Alexandra Chang
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$$\text{Let } X = |i - j|$$

- Total Possible Outcomes: $6 \times 6 = 36$
 - Largest value: $|1-6|$ or $|6-1| = 5$
 - Smallest value: $|1-1|, |2-2|, |3-3|, \dots = 0$
- ↓
- Possible values of $X \in \{0, 1, 2, 3, 4, 5\}$

• Probability Mass Function (PMF)

$X = i - j $	Pairs (i, j)	Count	PMF = $P(X=x)$
0	(1,1)(2,2)(3,3)(4,4)(5,5)(6,6)	6	$P(0) = 6/36$
1	(1,2)(2,1)(2,3)(3,2)(3,4)(4,3)(4,5)(5,4)(5,6)(6,5)	10	$P(1) = 10/36$
2	(1,3)(3,1)(2,4)(4,2)(3,5)(5,3)(4,6)(6,4)	8	$P(2) = 8/36$
3	(1,4)(4,1)(2,5)(5,2)(3,6)(6,3)	6	$P(3) = 6/36$
4	(1,5)(5,1)(2,6)(6,2)	4	$P(4) = 4/36$
5	(1,6)(6,1)	2	$P(5) = 2/36$

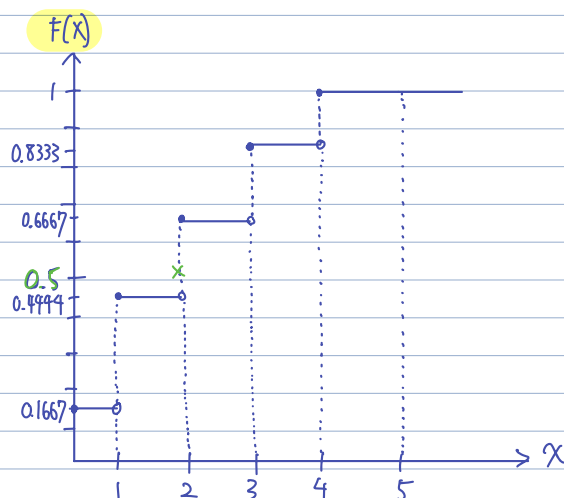
$$P(X=x) = \begin{cases} 6/36 & X=0 \\ 10/36 & X=1 \\ 8/36 & X=2 \\ 6/36 & X=3 \\ 4/36 & X=4 \\ 2/36 & X=5 \\ 0 & \text{otherwise} \end{cases}$$

• Cumulative Density Function (CDF)

$$F(x) = P(X \leq x)$$

X	$F(x)$	Interval
0	$6/36 = 0.1667$	$X \in [0, 1)$
1	$6/36 + 10/36 = 0.4444$	$X \in [1, 2)$
2	$6/36 + 10/36 + 8/36 = 0.6667$	$X \in [2, 3)$
3	$6/36 + 10/36 + 8/36 + 6/36 = 0.8333$	$X \in [3, 4)$
4	$6/36 + 10/36 + 8/36 + 6/36 + 4/36 = 0.9444$	$X \in [4, 5)$
5	$6/36 + 10/36 + 8/36 + 6/36 + 4/36 + 2/36 = 1$	$X \in [5, \infty)$

$$F(x) = \begin{cases} 0.1667 & 0 \leq X < 1 \\ 0.4444 & 1 \leq X < 2 \\ 0.6667 & 2 \leq X < 3 \\ 0.8333 & 3 \leq X < 4 \\ 0.9444 & 4 \leq X < 5 \\ 1 & X \geq 5 \end{cases}$$



$$\text{median} \\ P(X \leq m) \geq 0.5 \text{ and } P(X \geq m) \geq 0.5$$

$$P(X \leq 2) \geq 0.5 \text{ and } P(X \geq 2) \geq 0.5 \\ \text{median} = 2$$

The median is unique as 2 is the only value where this condition is satisfied.

2

Exercise 4.3. A motorist encounters four consecutive traffic lights, each equally likely to be red or green. Let X be the number of green lights passed by the motorist before being stopped by a red light. What is the pmf of X ?

Let X = # of G passed before stopping for R

- Total possible Outcomes: 5
- Largest value: 4
- Smallest value: 1

- R $X=0$
- GR $X=1$
- GGR $X=2$
- GGGR $X=3$
- GGGG $X=4$

↓

Possible values of $X \in \{0, 1, 2, 3, 4\}$

• Probability Mass Function (PMF)

X	$P(X)$
0	R = $\frac{1}{2} = 0.5$
1	GR = $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = 0.25$
2	GGR = $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} = 0.125$
3	GGGR = $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16} = 0.0625$
4	GGGG = $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16} = 0.0625$

$$P(X) = \begin{cases} 0.5 & X=0 \\ 0.25 & X=1 \\ 0.125 & X=2 \\ 0.0625 & X=3 \\ 0.0625 & X=4 \\ 0 & \text{otherwise} \end{cases}$$

3

Exercise 4.14. Suppose a fair die is tossed repeatedly until the first six occurs. Let X be the roll at which the first six is obtained. Find the expected value of X .

$$P(\text{rolling a } 6) = \frac{1}{6}$$

$$P(\text{not rolling a } 6) = \frac{5}{6}$$

X : Geometric Distribution with success probability: $p = \frac{1}{6}$

Tail Sum Formula

$$E(X) = \sum_{n=0}^{\infty} P(X > n)$$

- Nonnegative
- Integer
- RV

Geometric Distribution

X = number of trials until 1st success

- pmf: $P(X=k) = (1-p)^{k-1} p$, $k=1, 2, \dots$
- $E(X) = \frac{1}{p} = \frac{1}{1/6}$
- Variance: $1-p/p^2$

Step by step

$$P(X > 0): \text{1st "6" after roll } 0 = 1$$

$$P(X > 1): \text{1st "6" after roll } 1, \text{ aka 1st 6 happened when rolled for the 2nd time.}$$

$$\text{So probability of 1st roll with no 6} = \frac{5}{6}$$

$$P(X > 2): \text{So probability of 1st and 2nd roll with no 6} = \frac{5}{6} \cdot \frac{5}{6}$$

$$\left(\frac{5}{6}\right)^n$$

Probability we haven't seen a "6" in the first n rolls.

Compute Tail Probability

$$P(X > n) = (1-p)^n = \left(\frac{5}{6}\right)^n$$

Probability that 1st success happens after trial 'n'

Compute Expectation

$$E(X) = \sum_{n=0}^{\infty} \left(\frac{5}{6}\right)^n, \text{ infinite geometric with } \frac{1}{1-p} = \frac{1}{1/6}$$

$$E(X) = \frac{1}{p} = \frac{1}{1/6}$$

$E(X) = 6$ expected rolls until the first "6" appears.

4

Exercise 4.18. Suppose X can take only the values $-1, 0$, and 1 . If you want to make the mean of X zero and make the variance of X as large as possible, what must be the pmf of X ?

Given $E(X) = \mu = 0$

$\downarrow$
 $Var(X) = E(X^2)$

$$Var(X) = E(X^2) - \mu^2$$

Define pmf

$$\begin{aligned} & \bullet P(X=-1) = p_{-1} \\ & \bullet P(X=0) = p_0 \\ & \bullet P(X=1) = p_1 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} p_{-1} + p_0 + p_1 = 1$$

$$E(X) = \sum x_i p_i$$

Mean-zero Constraint

$$E(X) = -1(p_{-1}) + 0(p_0) + 1(p_1) = 0$$

Hence, in order to have $\mu = 0$, $p_{-1} = p_1$ with $p_0 = 0$

Maximize Variance

$$\begin{aligned} Var(X) = E(X^2) &= (-1)^2(p_{-1}) + (0)^2(p_0) + (1)^2(p_1) \\ &= p_{-1} + p_1 = 2p_{-1} = 2p_1 \end{aligned} \quad \Rightarrow \quad \begin{aligned} p_{-1} + p_0 + p_1 &= 1 \\ p_{-1} &= p_1 = 1/2 \end{aligned}$$

$$p(x) = \begin{cases} 1/2 & x = -1 \\ 0 & x = 0 \\ 1/2 & x = 1 \end{cases}$$

5

Exercise 4.20. Suppose a town has ten taxicabs, with license plate numbers $1, 2, \dots, 10$. You have observed the plate numbers of five of these ten cabs. Let X be the maximum of these five license plate numbers. Find the expected value and the variance of X . State your assumptions.

Assumptions:

- 5 license plates are drawn randomly without replacement from $\{1, 2, \dots, 10\}$
- All combinations of 5 cabs are equally likely. $\binom{10}{5} = 252$
- X is a discrete uniform order statistic representing the maximum of the sample.

Compute $P(X=x)$

forcing one to be the maximum, out of the remaining, choose 4

$$P(X=x) = \frac{\binom{x-1}{4}}{\binom{10}{5}}, \quad x = 5, 6, \dots, 10$$

Compute $E(X)$ and $Var(X)$

$$E(X) = \sum_{x=5}^{10} x \cdot P(X=x) = \sum_{x=5}^{10} x \cdot \frac{\binom{x-1}{4}}{\binom{10}{5}}$$

$$Var(X) = \sum_{x=5}^{10} x^2 \cdot P(X=x) = \sum_{x=5}^{10} x^2 \cdot \frac{\binom{x-1}{4}}{\binom{10}{5}}$$

X	$x \cdot \binom{x-1}{4}$	$x^2 \cdot \binom{x-1}{4}$
5	5	25
6	30	180
7	105	735
8	280	2240
9	630	5670
10	1260	12600
Sum	2310	21450

$$\binom{10}{5} = 252$$

$$E(X)^2 = 2310/252 = 9.17$$

$$Var(X) = 21450/252 = 85.12$$

$$Var(X) = E(X^2) - \mu^2$$

$$Var(X) = 85.12 - (9.17)^2 = 1.06$$

Homework 3

Stat-516

- Problem 4.2

The standard approach is to introduce a random variable $X = |Z_1 - Z_2|$ where Z_1 and Z_2 are the results of the first and second roll, respectively. It only remains to enumerate all possible values of X (from 0 to 5) with their corresponding probabilities.

- Problem 4.3

Note that the probability mass function (pmf) of the random variable X in question is a geometric distribution pmf with the probability of success (encountering a red light) being $p = \frac{1}{2}$. This implies that the probability $P(X = k) = (1 - p)^{k-1}p$ for a positive integer k .

- Problem 4.14

This is again a problem related to geometric distribution. More specifically, we have the random variable in question $X \sim Geo\left(\frac{1}{2}\right)$. Its expected value can be represented as

$$E(X) = \sum_{k=1}^{\infty} kp(1-p)^{k-1} = \frac{p}{1-p} \sum_{k=1}^{\infty} k(1-p)^k$$

To handle the last sum, it is useful to note that the derivative of $(1-p)^{k+1}$ with respect to p is exactly $(k+1)(1-p)^k = k(1-p)^k + (1-p)^k$. Adding and subtracting $(1-p)^k$ to each term in the above sum, one can interchange summation and differentiation in the sum above. Working with two simple geometric series sums, one ends up with the answer $\frac{1}{p}$.

- Problem 4.18

In this case, one can simply assume that the probability of taking the value of -1 is p_1 , that of 0 is p_2 , and the value of 1 is $1 - (p_1 + p_2)$. The resulting expectation will be $-p_1 + (1 - p_1 - p_2) = 1 - 2p_1 - p_2$. Selecting p_1 and p_2 such that $2p_1 + p_2 = 1$ gives the expectation being equal to zero. To find the variance, one has to find the $EX^2 = 1 - p_2$ first. Then, $V(X) = 1 - p_2 - (EX)^2 = 1 - p_2 - (1 - 2p_1 - p_2)^2$ which is a polynomial in p_1 and p_2 . It only remains to maximize this polynomial under constraint that $2p_1 + p_2$ is equal to 1.

- Problem 4.20

First, show that $P(X \leq x) = 0$ for $x < 5$ and is equal to $\frac{\binom{x}{5}}{\binom{10}{5}}$ when $x \geq 5$. From here, find the pmf of X .